

Lab 1: Implementing RBAC (Solution)

The screenshot shows the PostgreSQL IDE interface. On the left, the Object Explorer displays the database structure for 'dvdrental', including schemas like 'public' and tables like 'actor', 'address', 'category', 'city', 'country', 'customer', and 'film'. The 'film' table is selected, showing its columns, including 'film_id integer'. The main query editor on the right contains the following SQL code:

```
1 SET ROLE postgres;
2
3 CREATE ROLE cashier_role;
4
5 CREATE USER trang WITH PASSWORD 'trang123';
6 CREATE USER minh WITH PASSWORD 'minh123';
7
8 GRANT cashier_role TO trang;
9 GRANT cashier_role TO minh;
```

The 'Data Output' tab at the bottom shows the result of the query: 'GRANT ROLE' and 'Query returned successfully in 38 msec.' The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.038' and 'LF Ln 9, Col 28'.

The screenshot shows the PostgreSQL IDE interface with the same database structure as the previous screenshot. The main query editor on the right contains the following SQL code:

```
1 SET ROLE postgres;
2
3 CREATE ROLE cashier_role;
4
5 CREATE USER trang WITH PASSWORD 'trang123';
6 CREATE USER minh WITH PASSWORD 'minh123';
7
8 GRANT cashier_role TO trang;
9 GRANT cashier_role TO minh;
10
11 GRANT SELECT ON film, staff TO cashier_role;
12 GRANT SELECT, INSERT ON payment TO cashier_role;
```

The 'Data Output' tab at the bottom shows the result of the query: 'GRANT' and 'Query returned successfully in 38 msec.' The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.030' and 'LF Ln 12, Col 50'. A green notification bar at the bottom right states 'Query returned successfully in 30 msec.'

Lab 2: Implementing RLS (Task)

Object Explorer

- Servers (1)
 - PostgreSQL 18
 - Databases (2)
 - dvdrental
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (15)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - film
 - Columns (13)
 - film_id integer

Query

```
1 CREATE ROLE cashier_role;
2
3 CREATE USER trang WITH PASSWORD 'trang123';
4 CREATE USER minh WITH PASSWORD 'minh123';
5
6 GRANT cashier_role TO trang;
7 GRANT cashier_role TO minh;
8
9 GRANT SELECT ON film, staff TO cashier_role;
10 GRANT SELECT , INSERT ON payment TO cashier_role;
11
12 ALTER TABLE payment ENABLE ROW LEVEL SECURITY ;
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 47 msec.

Total rows: Query complete 00:00:00.047 LF Ln 12, Col 48

Object Explorer

- Servers (1)
 - PostgreSQL 18
 - Databases (2)
 - dvdrental
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (15)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - film
 - Columns (13)
 - film_id integer

Query

```
5
6 GRANT cashier_role TO trang;
7 GRANT cashier_role TO minh;
8
9 GRANT SELECT ON film, staff TO cashier_role;
10 GRANT SELECT , INSERT ON payment TO cashier_role;
11
12 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
13
14 CREATE POLICY isolate_cashier_payments ON payment
15 FOR SELECT
16 TO cashier_role
17 USING (
18     staff_id = ( SELECT staff_id FROM staff WHERE username = current_user )
19 );
```

Data Output Messages Notifications

CREATE POLICY

Query returned successfully in 30 msec.

✓ Query returned successfully in 30 msec. ✕

Total rows: Query complete 00:00:00.030 LF Ln 19, Col 3

Object Explorer

Servers (1)

PostgreSQL 18

Databases (2)

dvdrental

public

Aggregates

Collations

Domains

FTS Configurations

FTS Dictionaries

FTS Parsers

FTS Templates

Foreign Tables

Functions

Materialized Views

Operators

Procedures

Sequences

Tables (15)

actor

address

category

city

country

customer

film

Columns (13)

film_id integer

dvdrental/postgres@PostgreSQL 18*

Query

```
10 GRANT SELECT , INSERT ON payment TO minh;
11
12 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
13
14 CREATE POLICY isolate_cashier_payments ON payment
15 FOR SELECT
16 TO cashier_role
17 USING (
18     staff_id = ( SELECT staff_id FROM staff WHERE username = current_user )
19 );
20
21 SET ROLE trang;
22 SELECT COUNT(*) FROM payment;
23
24 SET ROLE minh;
```

Query History

Execute script

Data Output

Messages

Notifications

SET

Query returned successfully in 41 msec.

Total rows: Query complete 00:00:00.041

LF Ln 24, Col 15

Object Explorer

Servers (1)

PostgreSQL 18

Databases (2)

dvdrental

public

Aggregates

Collations

Domains

FTS Configurations

FTS Dictionaries

FTS Parsers

FTS Templates

Foreign Tables

Functions

Materialized Views

Operators

Procedures

Sequences

Tables (15)

actor

address

category

city

country

customer

film

Columns (13)

film_id integer

dvdrental/postgres@PostgreSQL 18*

Query

```
11 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
12
13 CREATE POLICY isolate_cashier_payments ON payment
14 FOR SELECT
15 TO cashier_role
16 USING (
17     staff_id = ( SELECT staff_id FROM staff WHERE username = current_user )
18 );
19
20 SET ROLE trang;
21 SELECT COUNT(*) FROM payment;
22
23 SET ROLE minh;
24 SELECT COUNT(*) FROM payment;
```

Query History

Execute script

Data Output

Messages

Notifications

Showing rows: 1 to 1

Page No: 1 of 1

count
0

Successfully run. Total query runtime: 32 msec. 1 rows affected.

Total rows: 1 Query complete 00:00:00.032

LF Ln 25, Col 30